

Data and Network Security

(Master Degree in Computer Science and Cybersecurity)

Lecture 2
Spring 2026



Outline for today

- Recap previous lecture
- Malware types
- Emerging threats

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- **Recap previous lecture**
- Malware types
- Emerging threats

Cybersecurity / Data and Network Security?

Cybersecurity is the practice of protecting digital systems, networks, and data from unauthorized access, alteration, or destruction. It encompasses various technologies, processes, and practices designed to safeguard information assets against a wide range of cyber threats.

The goals of DNS



Confidentiality

Protecting sensitive information
from unauthorized disclosure.



Integrity

Ensuring the accuracy and trustworthiness of data by preventing unauthorized modifications

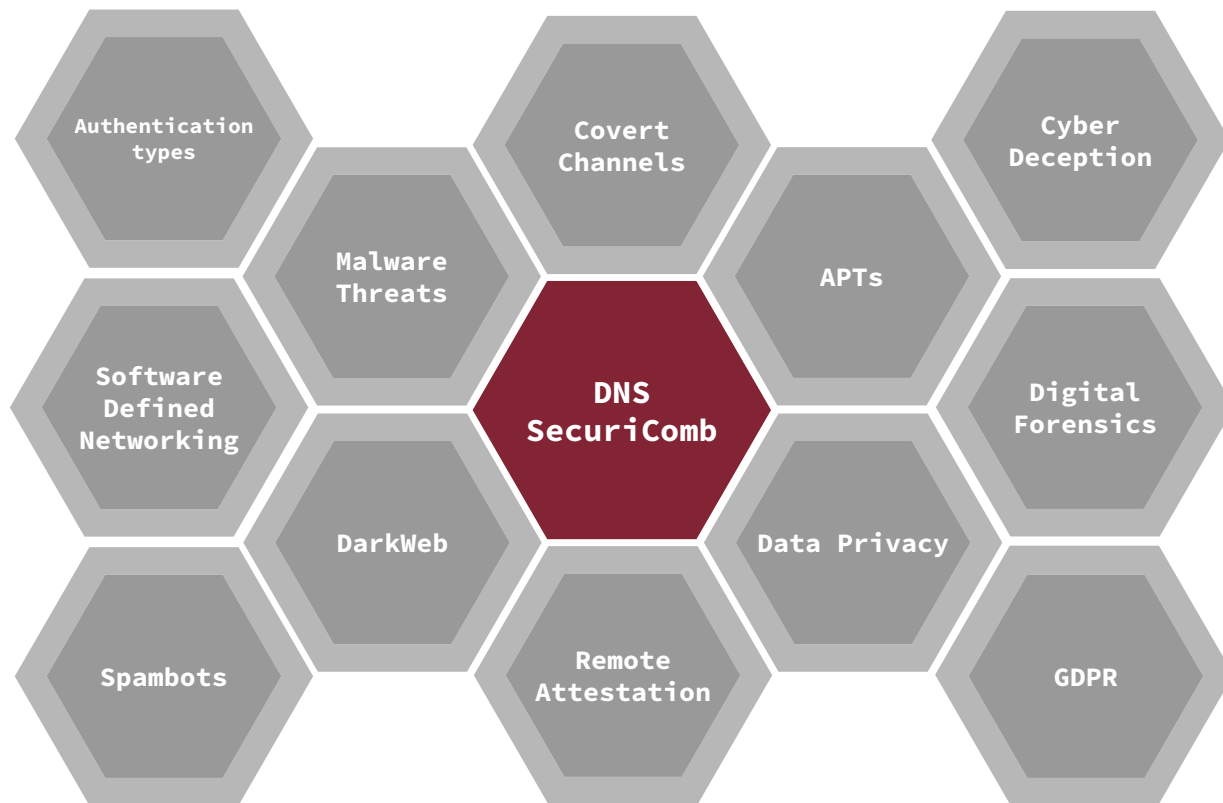


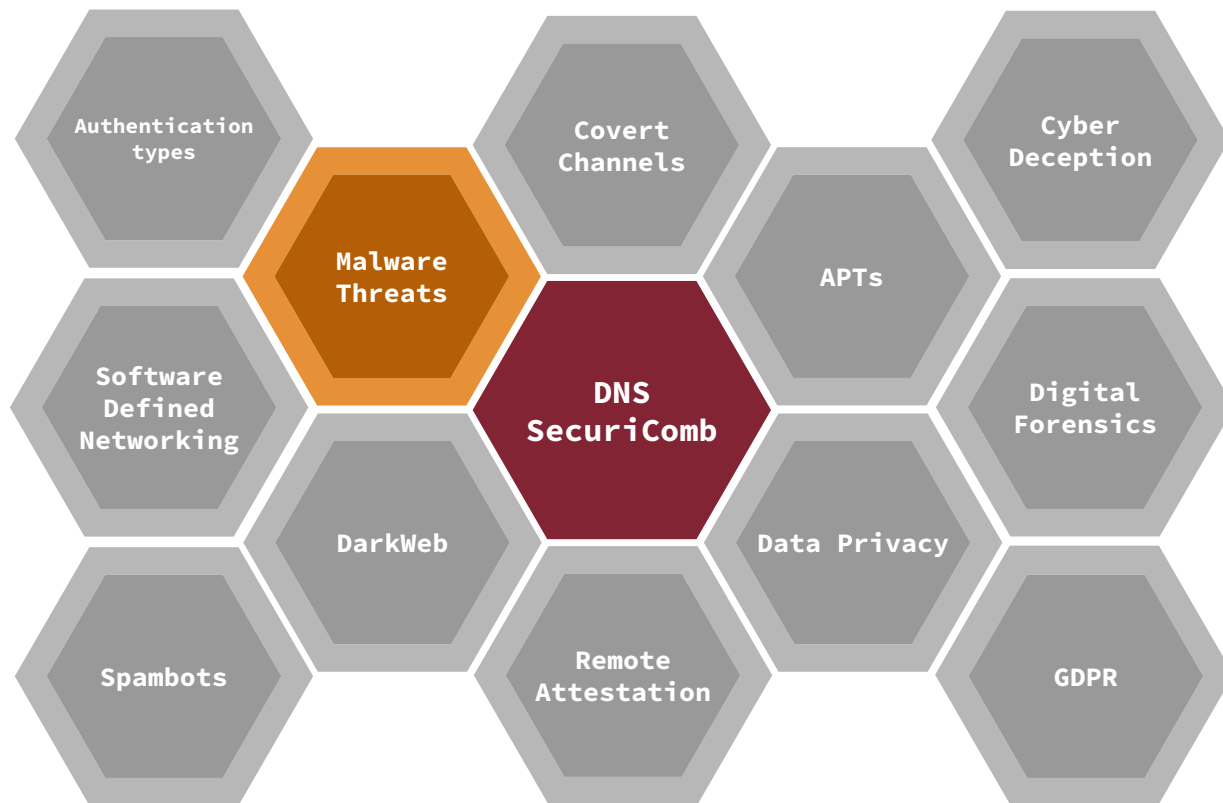
Availability



Ensuring that data and resources are available and accessible to authorized users when needed.

Major Threats to DATAs CIA





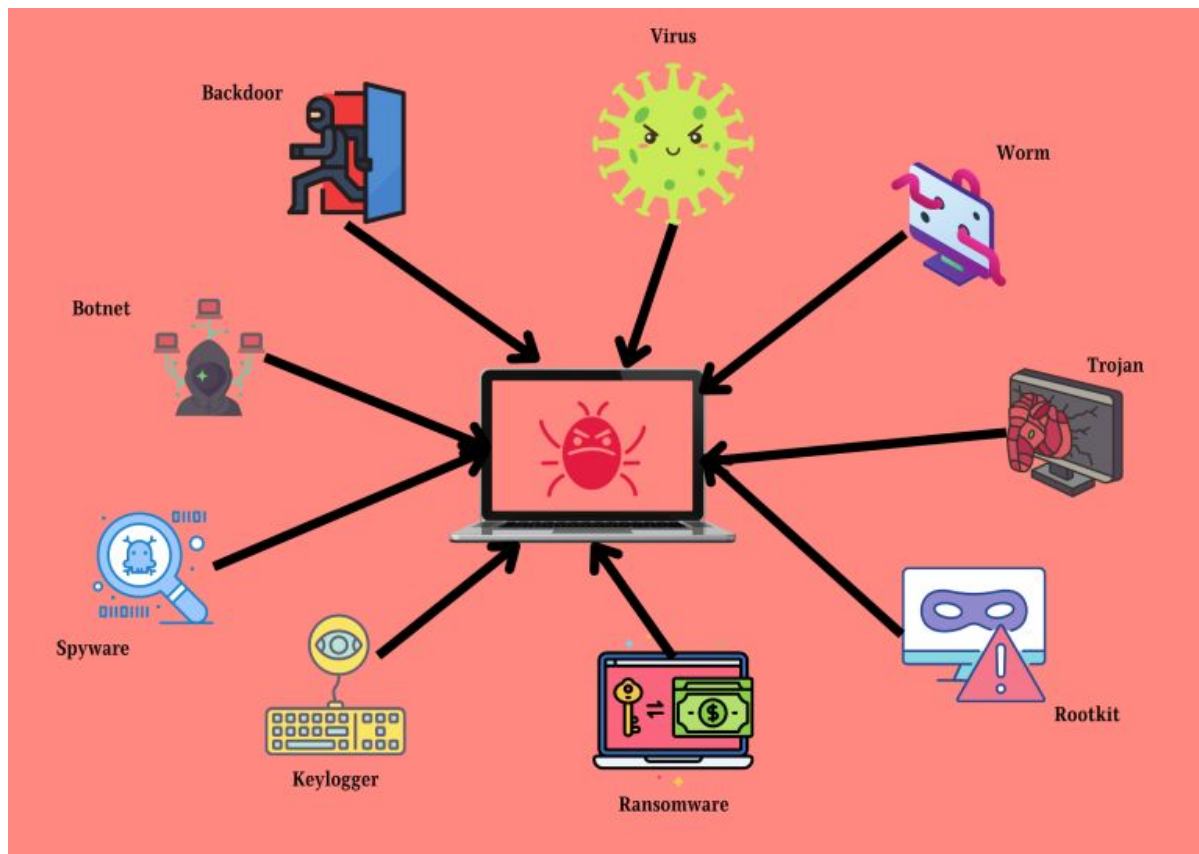
Malware Threats

Malware:

Type of software program or code specifically designed to infiltrate, damage, disrupt, or gain unauthorized access to computer systems, networks, or devices, often with malicious intent.

Broad category that encompasses various types of malicious programs, each with its own specific behavior and objectives.

Malware Types



Malware Threats - knowing them

1. Identification and Detection
2. Prevention and mitigation
3. Remediation
4. Risk (assessment and management)
5. General user education
6. Adapting towards evolving threats

Malware Threats - knowing them

1. Identification and Detection

By recognizing the signs of malicious activity, such as unusual network activity or unauthorized system changes, security teams can respond promptly to mitigate potential risks.



Malware Threats - knowing them

1. Identification and Detection
- 2. Prevention and mitigation**
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Malware Threats - knowing them

2. Prevention and mitigation

Understanding how malware operates enables organizations to implement **proactive** measures to prevent and mitigate malware.

Employing security controls such as **antivirus**, **firewalls**, **IDS**, and secure configuration practices, organizations can reduce the likelihood of malware attacks and limit their impact.



Malware Threats - knowing them

1. Identification and Detection
2. Prevention and mitigation
- 3. Remediation**
4. Risk (assessment and management)
5. General user education
6. Adapting towards evolving threats

Malware Threats - knowing them

3. Remediation

Knowing malware types and how they operate facilitates an effective response and remediation process.

Security teams can leverage their understanding of specific malware behaviors to contain infections, remove malicious code, and restore affected systems to a secure state.



Malware Threats - knowing them

1. Identification and Detection
2. Prevention and mitigation
3. Remediation
- 4. Risk (assessment and management)**
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Malware Threats - knowing them

4. Risk (assessment and management)

Identifying potential threats and vulnerabilities associated with different malware, entities can prioritize security measures and allocate resources to address the most significant risks.



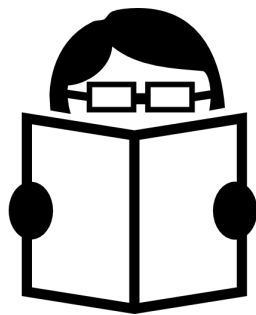
Malware Threats - knowing them

1. Identification and Detection
2. Prevention and mitigation
3. Remediation
4. Risk (assessment and management)
- 5. General user education**
6. Adapting towards evolving threats

Malware Threats - knowing them

5. General user education

Teaching users to recognize common signs of malware attacks, such as suspicious emails or unexpected pop-up messages, organizations can assist them in taking proactive measures to protect themselves and the organization.



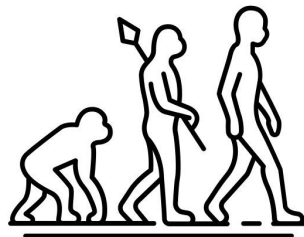
Malware Threats - knowing them

1. Identification and Detection
2. Prevention and mitigation
3. Remediation
4. Risk (assessment and management)
5. General user education
6. **Adapting towards evolving threats**

Malware Threats - knowing them

6. Adapting towards evolving threats

Continuously monitoring and analyzing emerging malware trends, organizations can enhance their cybersecurity position and try to stay one step ahead of adversaries.

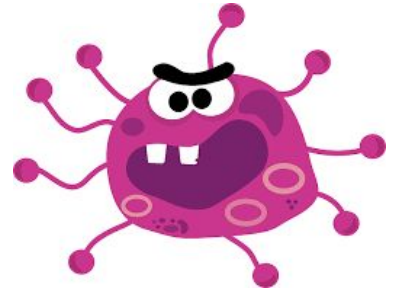


Malware Types - Common

- Viruses
- Worms
- Trojans
- Ransomware
- Spyware
- Adware
- Rootkits
- Scareware

Virus

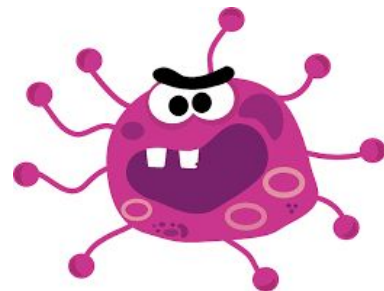
Program that can infect other programs by modifying them to include a, possibly evolved, version of itself, with intent to cause damage.



Virus

Program that can infect other programs by modifying them to include a, possibly evolved, version of itself, with intent to cause damage.

ILOVEYOU (discovered in 2000). The malware was delivered to millions of users as an email attachment with the subject line “ILOVEYOU.” Once opened, it spread to every contact in a user’s Microsoft Outlook address book and overwrite certain files (e.g., JPEG and MP3 files) from the hard drive.



Trojan

Class of malware that appears to perform a desirable function but in fact performs undisclosed malicious functions that usually allow unauthorized access to the victim computer.



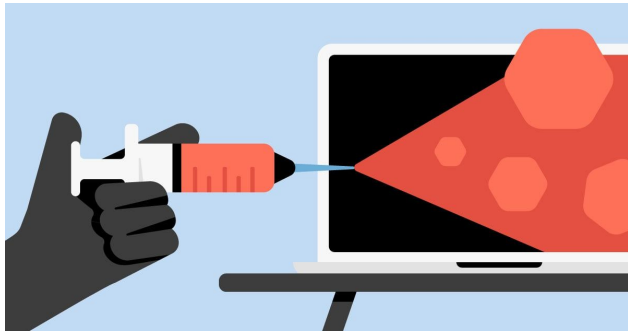
Ransomware



Type of malware from cryptovirology that threatens to publish the victim's data or perpetually block access to it unless a ransom is paid.

Rootkit

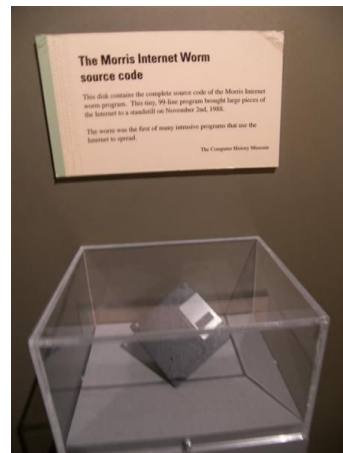
A rootkit is a program or a collection of malicious software tools that give an adversary remote access to and control over a computer.



Worms



A computer worm is a type of malware that can automatically propagate or self-replicate without human interaction, enabling its spread to other computers across a network.



Spyware

Malicious software that infects PCs and mobile devices in order to collect information on users and data on browsing habits, internet use, etc.



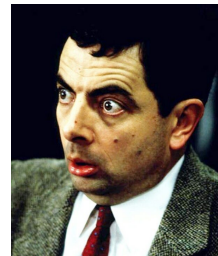
Adware

Adware is a type of malicious software that secretly installs itself on your device and displays unwanted advertisements and pop-ups. In some cases, adware can even track your online behavior and display personalized ads.



Scareware

Malware that scares people into visiting spoofed or infected websites or downloading malicious software. Scareware can come in the form of pop-up ads that appear on a user's computer or spread through spam email attacks.



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Ransomware



Type of malware from cryptovirology that threatens to publish the victim's data or perpetually block access to it unless a ransom is paid.

The Ransomware Threat

NHS cyber-attack: GPs and hospitals hit by ransomware

🕒 13 May 2017

[f](#) [💬](#) [🐦](#) [✉](#) [Share](#)

Worldwide ransomware hack hits hospitals, phone companies

The ransomware attack has hit 16 NHS hospitals in the UK and up to 70,000 devices across 74 countries using a leaked exploit first discovered by the NSA.



Alfred Ng [👤](#) May 14, 2017 10:20 AM PDT

NEWS

Ransomware attack hits North Carolina water utility following hurricane

A North Carolina water utility still recovering from Hurricane Florence became the victim of a ransomware attack.

[🐦](#) [f](#) [in](#) [🍷](#) [📧](#) [📺](#)

Colonial Pipeline hack explained: Everything you need to know

A ransomware attack brought a major gas pipeline to a standstill in May. Here's what happened and who was behind the hack.



By **Sean Michael Kerner**

Published: 26 Apr 2022

Signature based vs. Behaviour based detection



Signature based vs. Behaviour based detection



Signature based detection works by searching for a known identity – or signature – for each specific event.

- Very efficient (as long as it is kept up to date)

Signature based vs. Behaviour based detection

Analysing and monitoring how a process behaves in the system, for example how many files it accesses, what locations of the storage affects etc.



Ransomware detectors

ShieldFS: A Self-healing, Ransomware-aware Filesystem

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DEIB, Politecnico di Milano, Milan, Italy

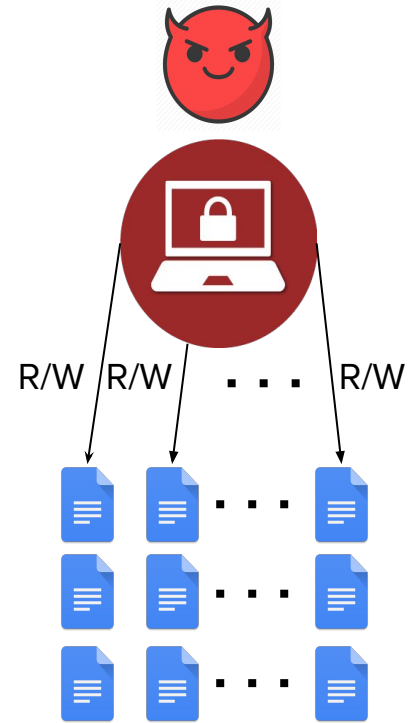
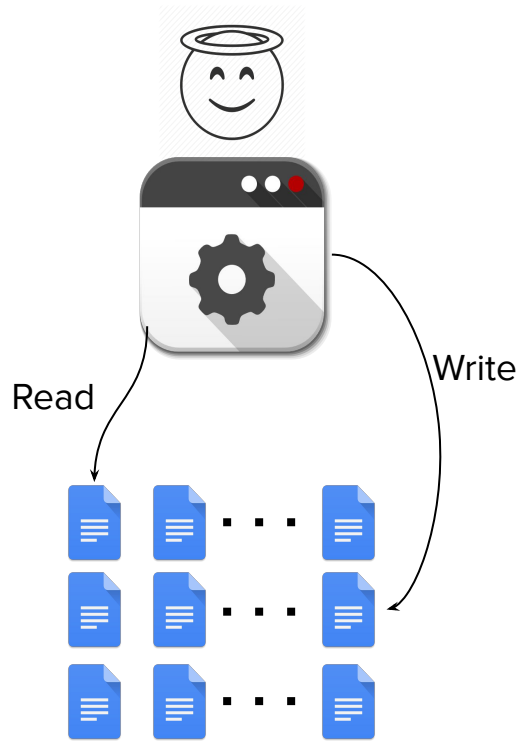


RWGuard: A Real-Time Detection System Against Cryptographic Ransomware

Shagufta Mehnaz^(✉), Anand Mudgerikar, and Elisa Bertino

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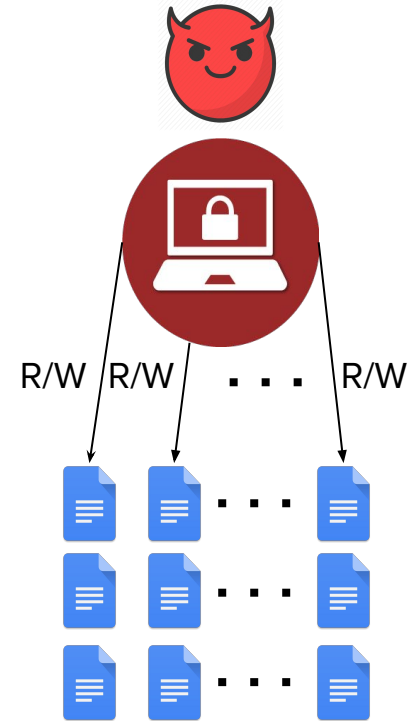
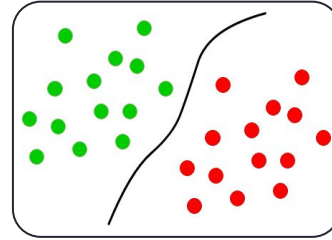
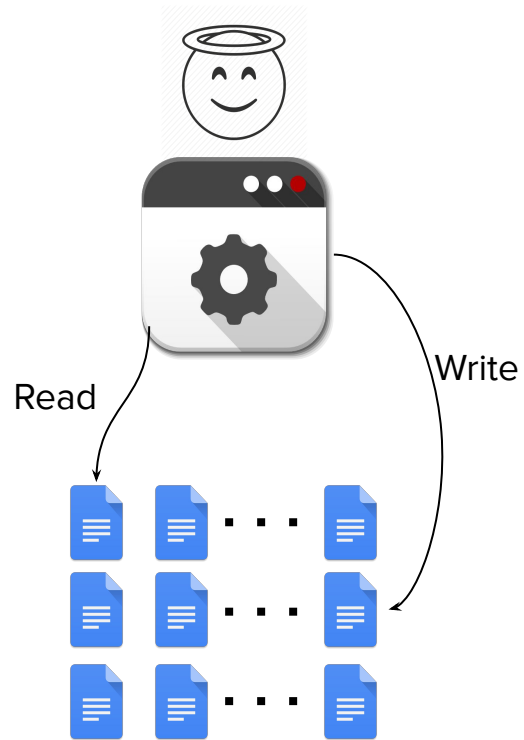
Ransomware behaviour



Ransomware features

- **Encrypts files** -> - high entropy
 - overwrites whole file
 - completely changes file content (no similarity)
 - changes file type
- **Access as many files as possible** -> lots of listing/read/write/open/create/close
- **Encrypt all user files** -> - access different, unrelated file types
 - access all files in every directory
- **Encrypts as fast as possible** -> very high access frequency

Ransomware detectors



Behavioural Classification

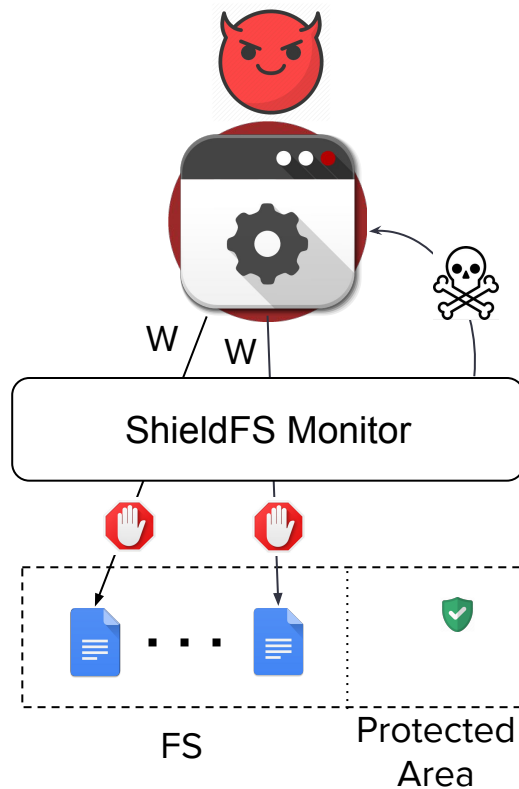
Behavioural classifiers analyse features inextricably linked with ransomware

- e.g., high number of read/write/directory listing, high entropy writes**

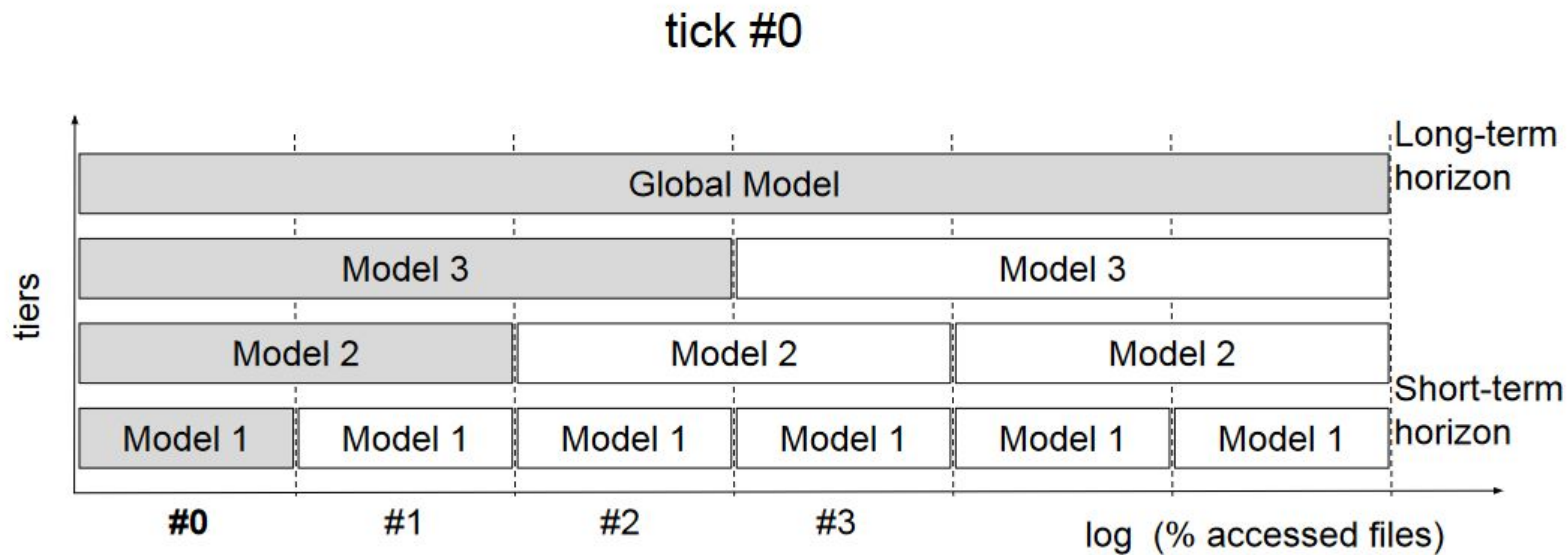
Model behavior of individual processes

- per-process feature collection**

ShieldFS by Continella et al.

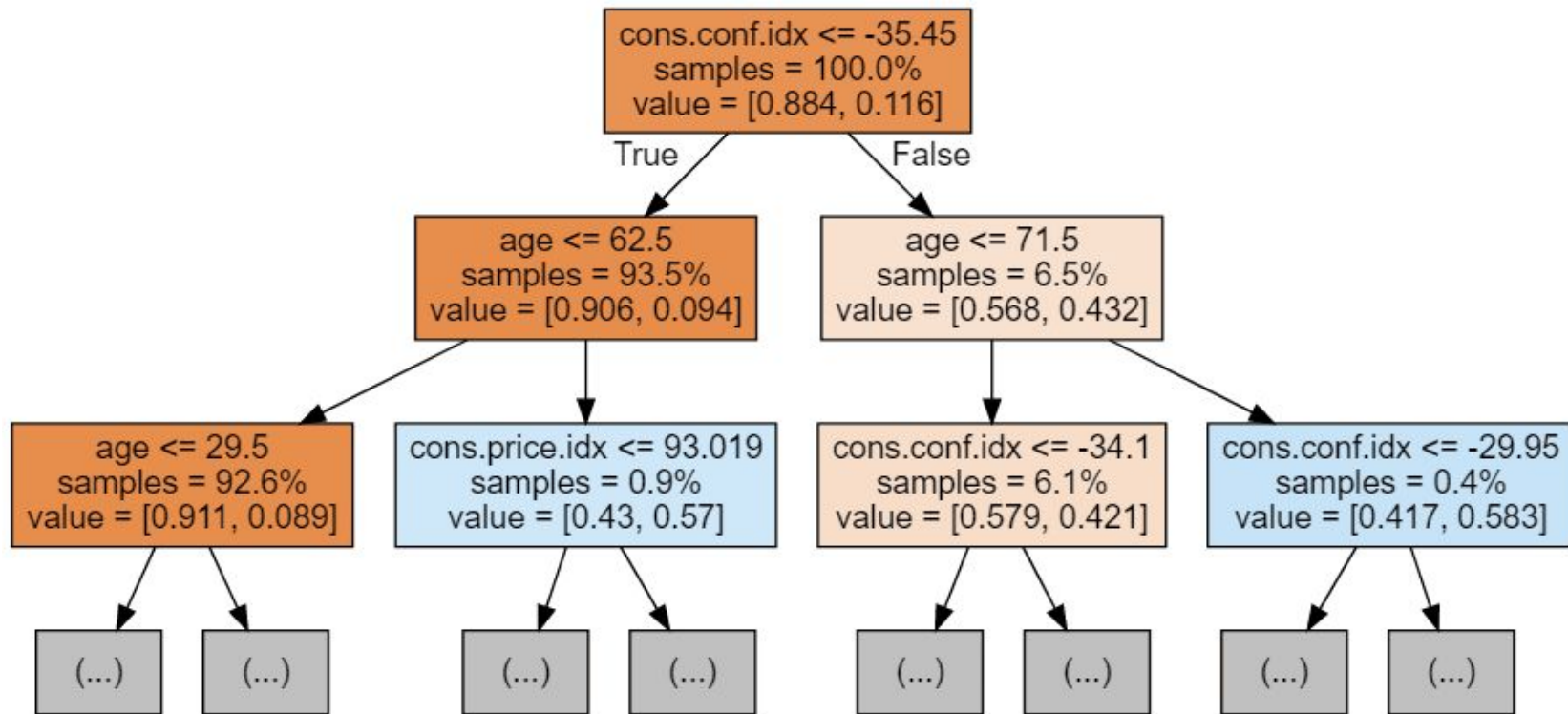


ShieldFS Detector



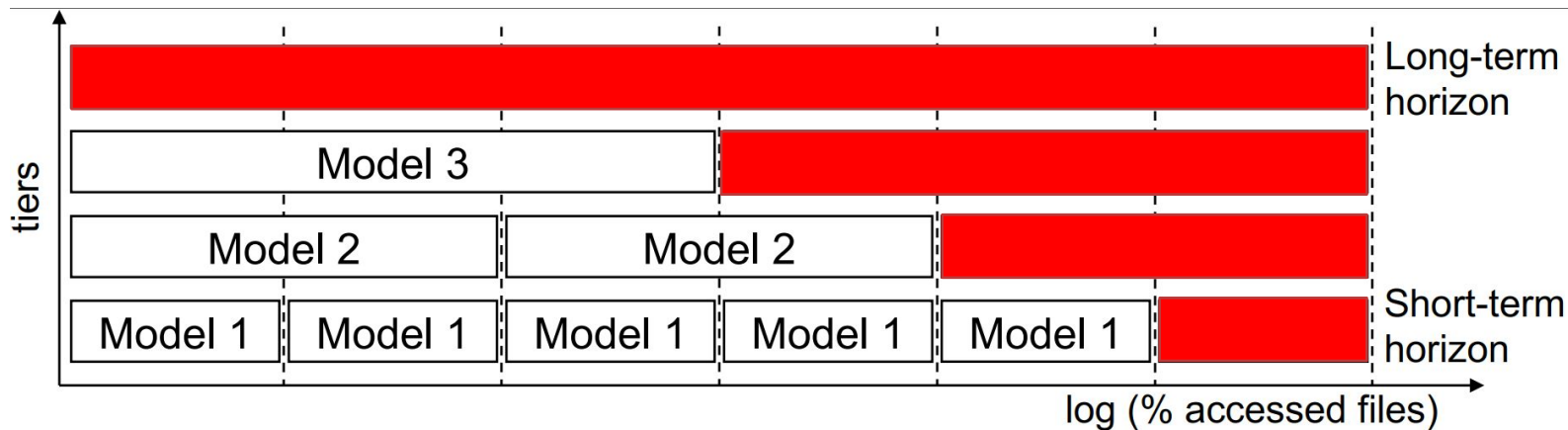
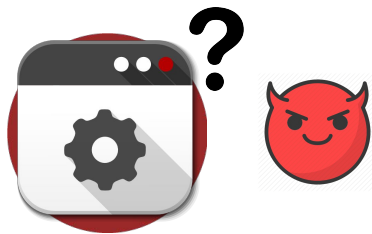
ShieldFS Detector - Random Forests?

$\langle f_1, f_2, \dots, f_N \rangle$

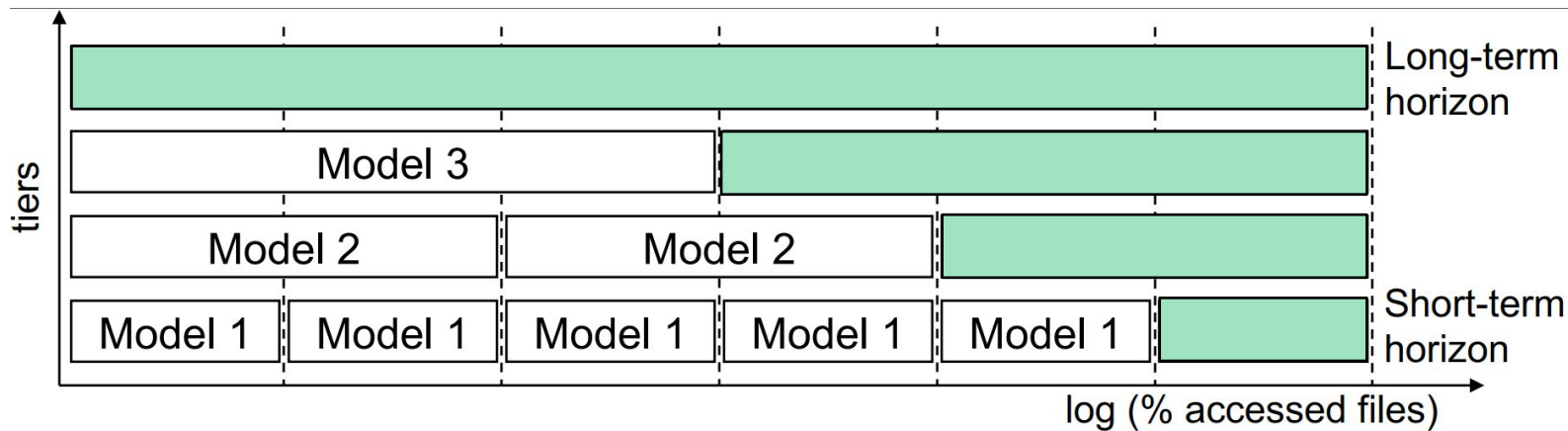
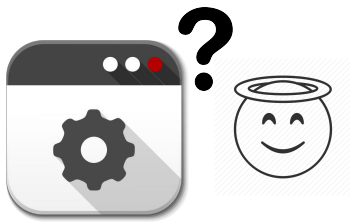


*illustrative example

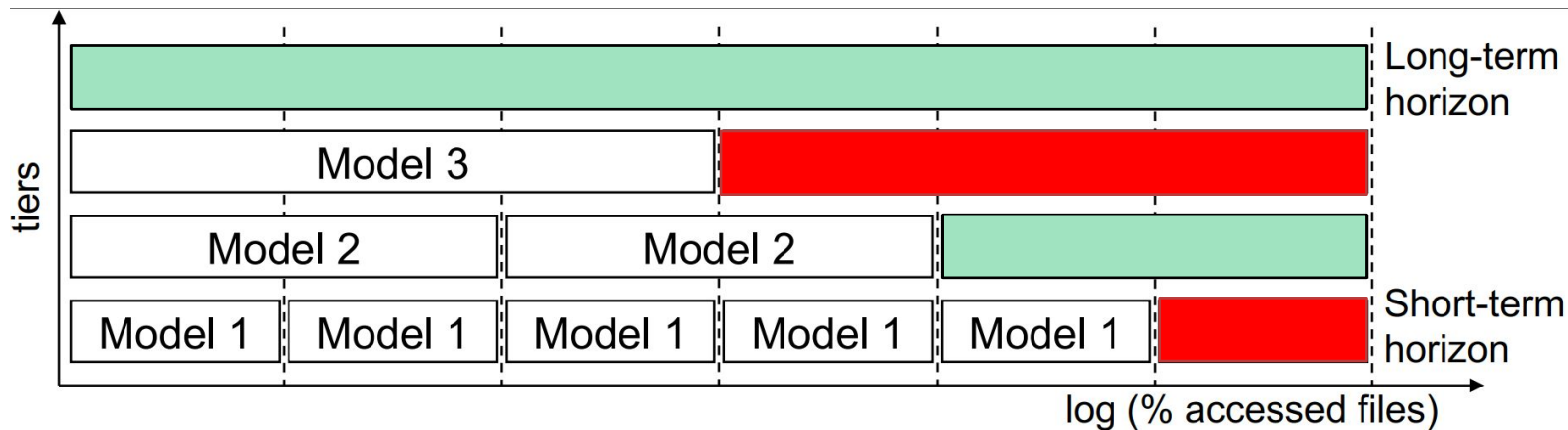
ShieldFS Detection Process



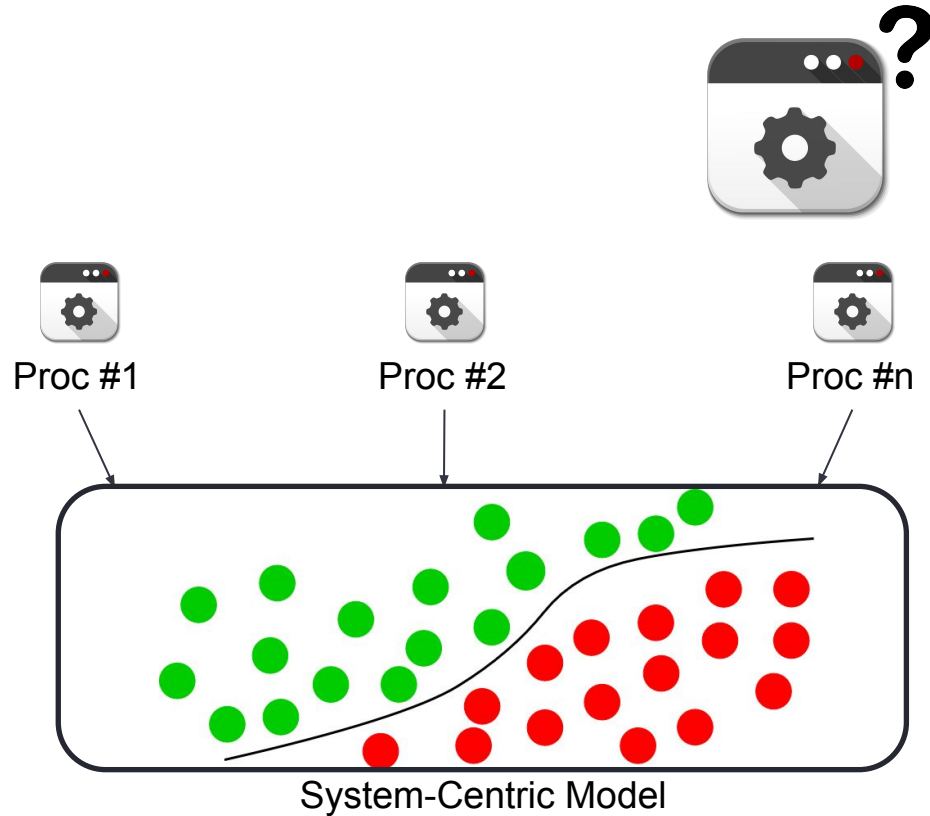
ShieldFS Detection Process



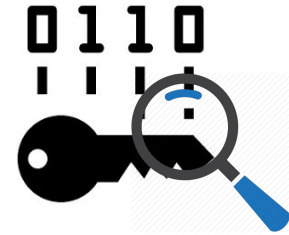
ShieldFS Detection Process



ShieldFS Detection Process

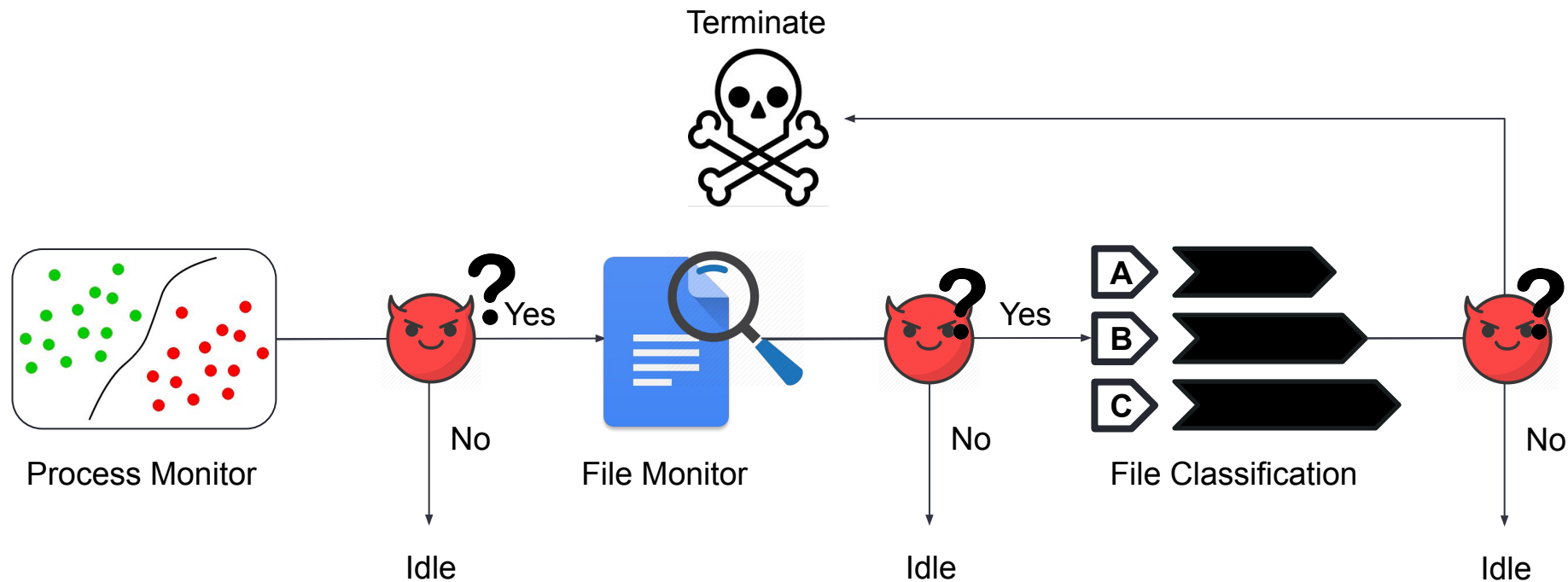


+



Search for Crypto Functions

RWGuard by Mehnaz et al.



Are these approaches reliable in adversarial conditions?



Evading Behavioural Classification

How can we lower the expression of all ransomware features at the process level?

- Reduce feature expression by reducing #operations**

Evading Behavioural Classification

How can we lower the expression of all ransomware features at the process level?

- Reduce feature expression by reducing #operations

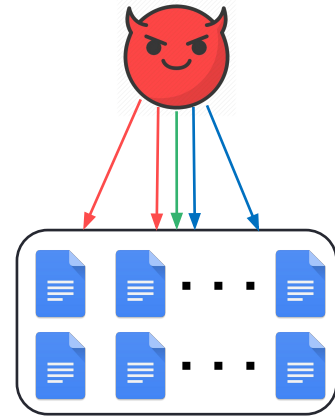
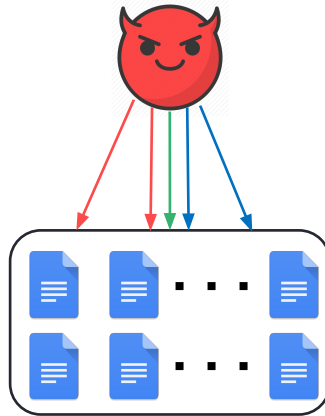
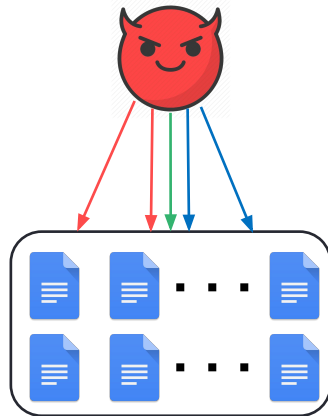
Distribute ransomware operations over independent, cooperating processes

- Process Splitting
- Functional Splitting
- Mimicry

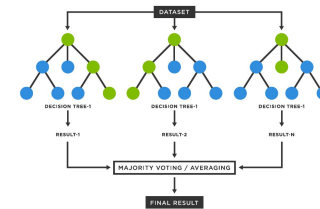
Process splitting

Process Splitting

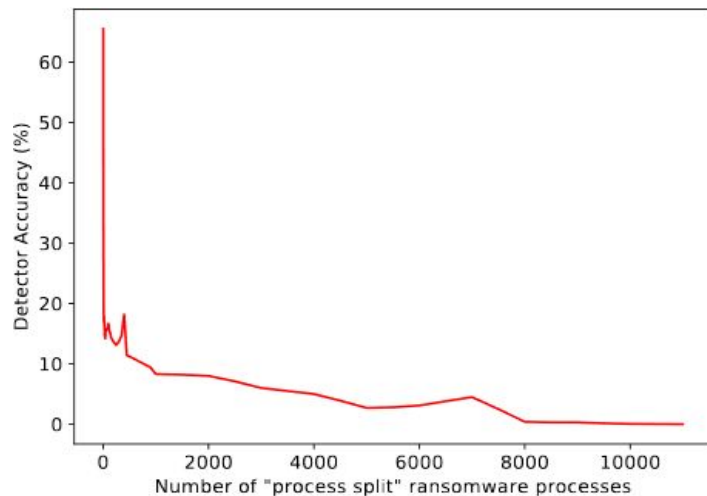
- Ransomware function 1
- Ransomware function 2
- Ransomware function 3



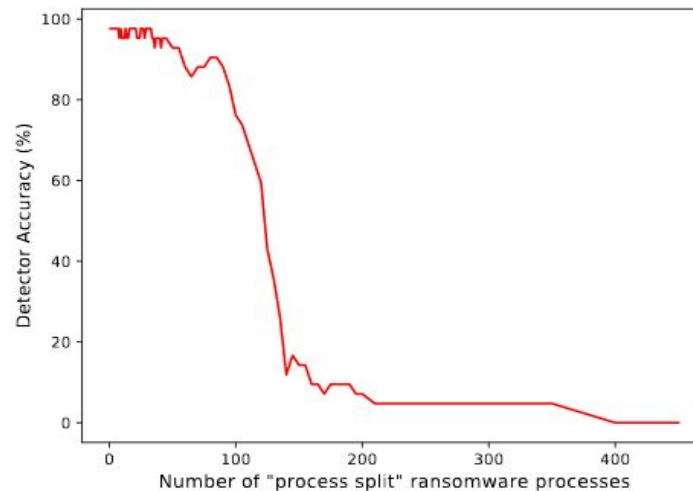
Process Splitting Evaluation



ShieldFS



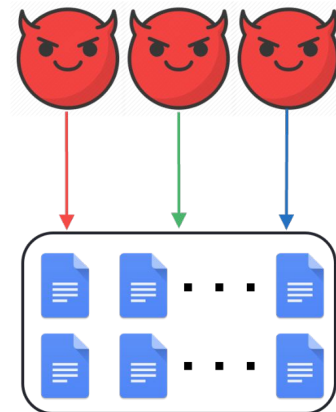
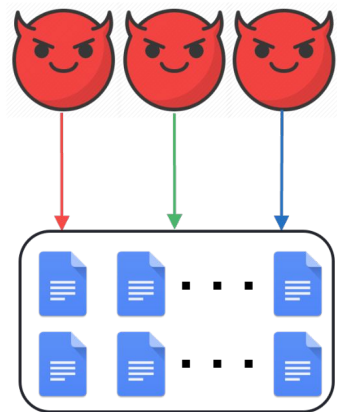
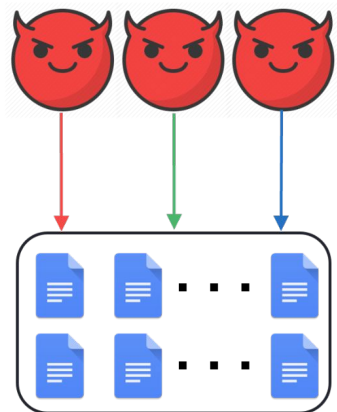
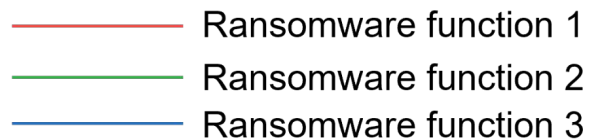
RWGuard



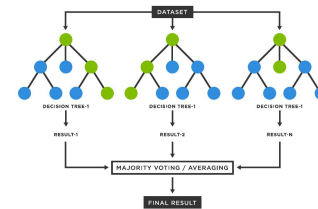
Functional splitting

Functional

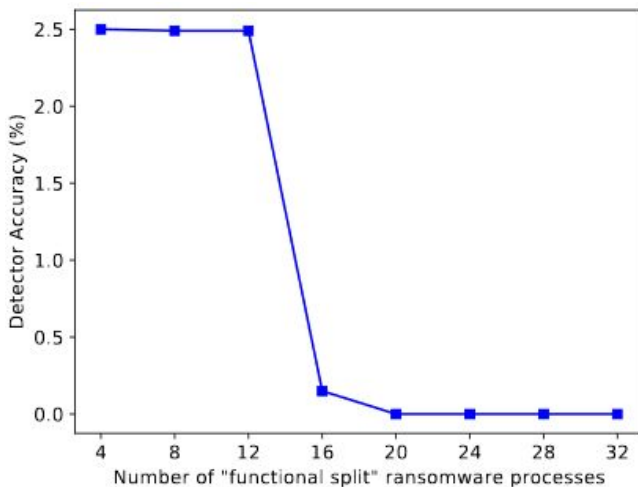
Splitting



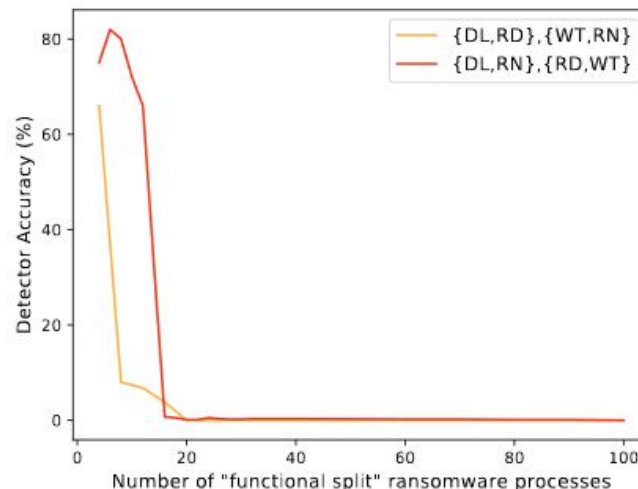
Functional Splitting Evaluation



ShieldFS

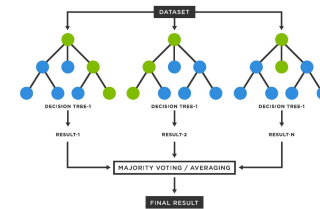


(a) Single functional splitting

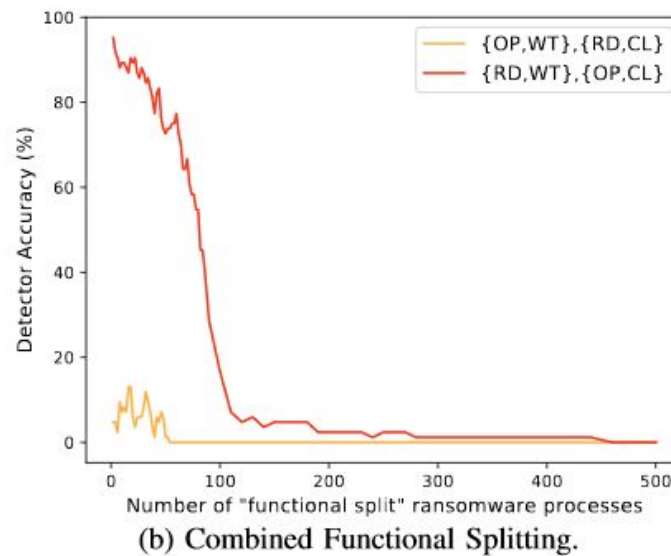
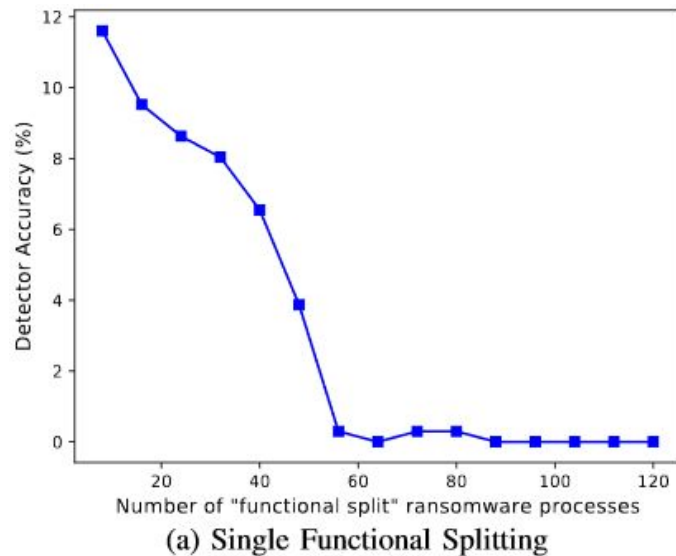


(b) Combined Functional Splitting

Functional Splitting Evaluation

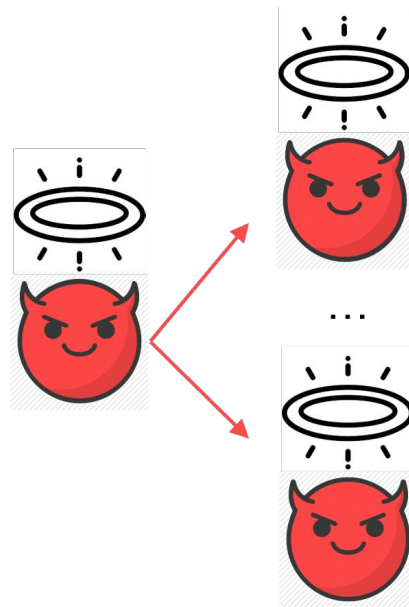
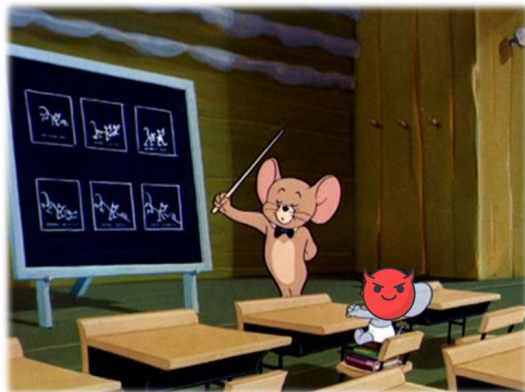


RWGuard



Mimicry

Mimicry:



Mimicry Evaluation

ShieldFS: full evasion

- 170 mimicry processes

RWGuard: full evasion

- 170 mimicry processes

Malwarebytes: full evasion

- 470 mimicry processes

Ransomware detectors

ShieldFS: A Self-healing, Ransomware-aware Filesystem

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RWGuard: A Real-Time Detection System Against Cryptographic Ransomware

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Purdue University, West Lafayette, IN, USA
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Malwarebytes

Can we make these approaches more reliable?



A naive approach

- Update the behavioural classifiers on these workload distribution attacks

A naive approach

- Update the behavioural classifiers on these workload distribution attacks
- works on process splitting and functional splitting

A naive approach

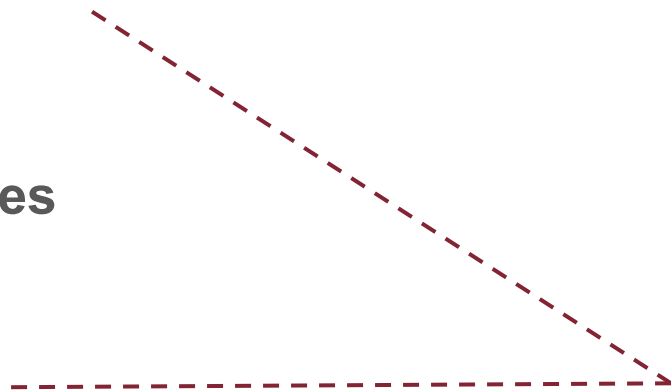
- Update the behavioural classifiers on these workload distribution attacks
 - works on process splitting and functional splitting
 - But what about Mimicry?



Will shifting focus help?



Process-Level Features

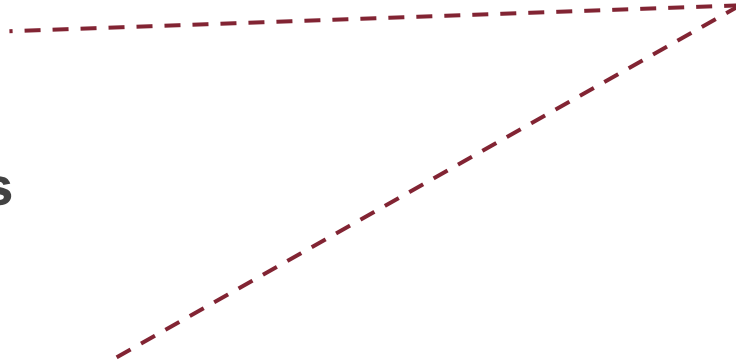


File-Level Features



Process-Level Features

File-Level Features



Detection Components

- Disk activity monitor
- File based behavioral detector
- File recovery module

Detection Components

- Disk activity monitor
- **File based behavioral detector**
- File recovery module

File-level features

- Read/Write data mismatch
- File write ratio
- File read ratio
- Number of Processes Reading or Writing the File
- Number of Operations on the File

File-level features

- **Read/Write data mismatch**
- File write ratio
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File-level features

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File-level features

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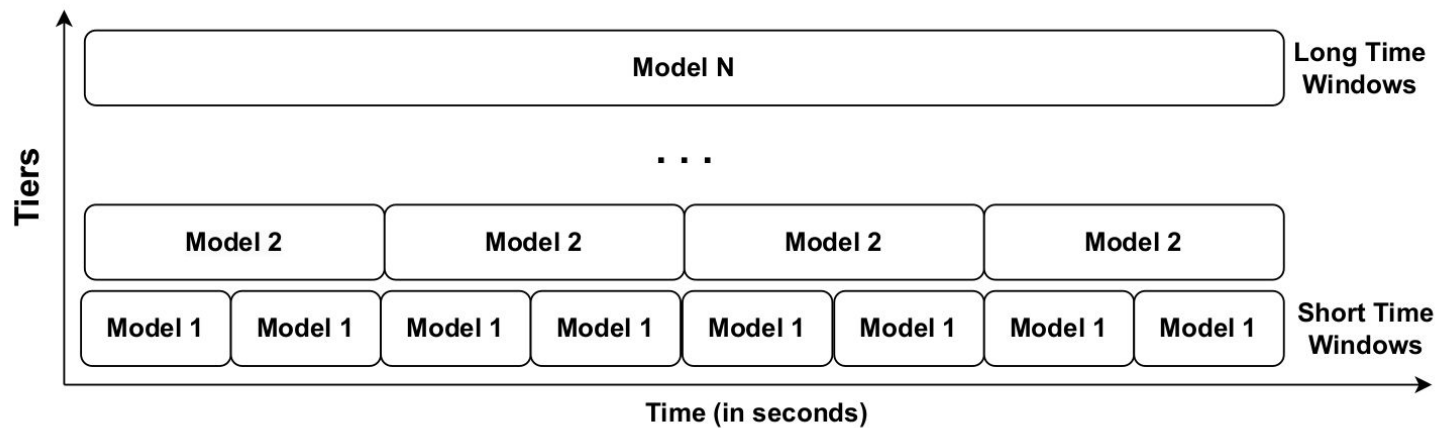
File-level features

- Read/Write data mismatch
- File write ratio
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- **Number of Processes Reading or Writing the File**
- Number of Operations on the File

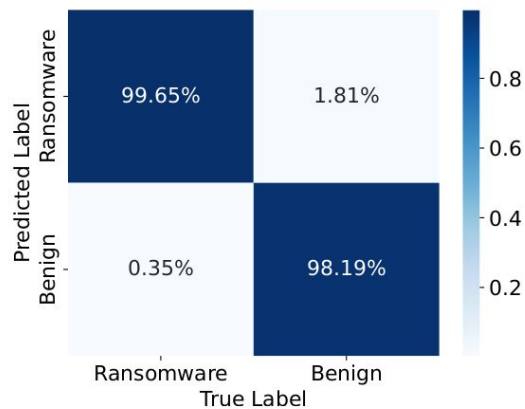
File-level features

- Read/Write data mismatch
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- Number of Processes Reading or Writing the File
- **Number of Operations on the File**

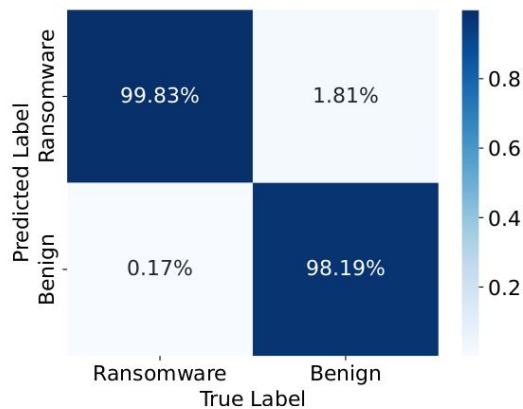
Detectors - ShieldFS inspired



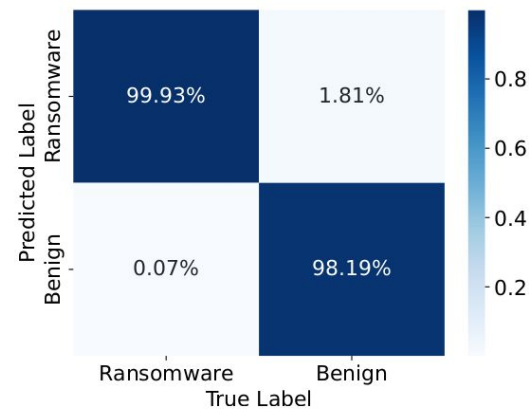
File-centric: How does it perform?



(a) Benign vs. Traditional Ransomware



(b) Benign vs. Evasive Ransomware



(c) Benign vs. Adaptive Ransomware

Reading Material

1. Behavioral ransomware detectors: Attached to this lecture in the Classwork section.

NOTE: All reading material are included in the oral examination unless specified otherwise.